

GEMP ACCESS COURSE: BIOLOGICAL CHEMISTRY

Lecturer: Christopher Larbie, *PhD*

Lecture 5: Alkanes

Kwame Nkrumah University of Science and Technology, Kumasi | Leaders In Change

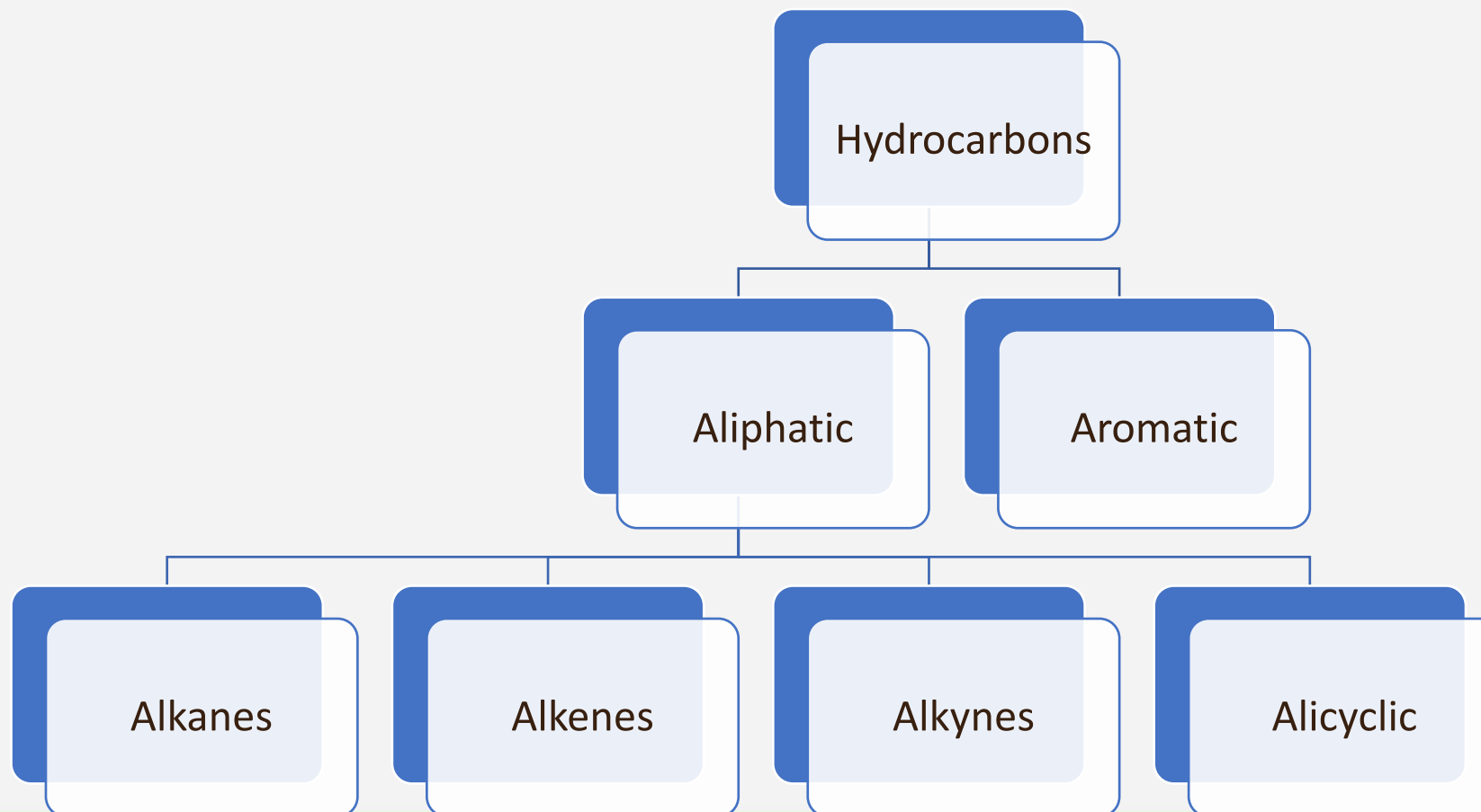
Visit us at  www.knust.edu.gh

 uro@knust.edu.gh

Follow KNUST on:



HYDROCARBONS

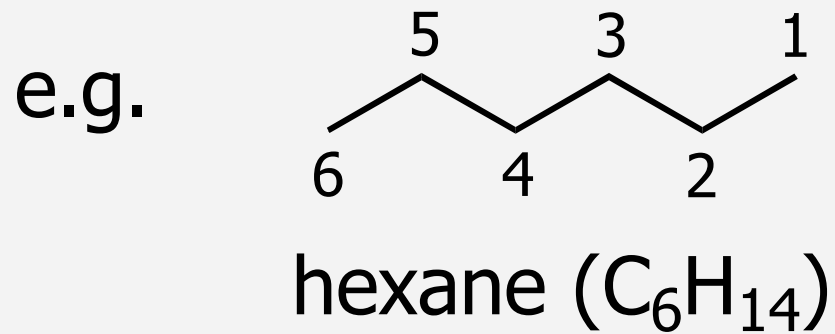


ALKANES

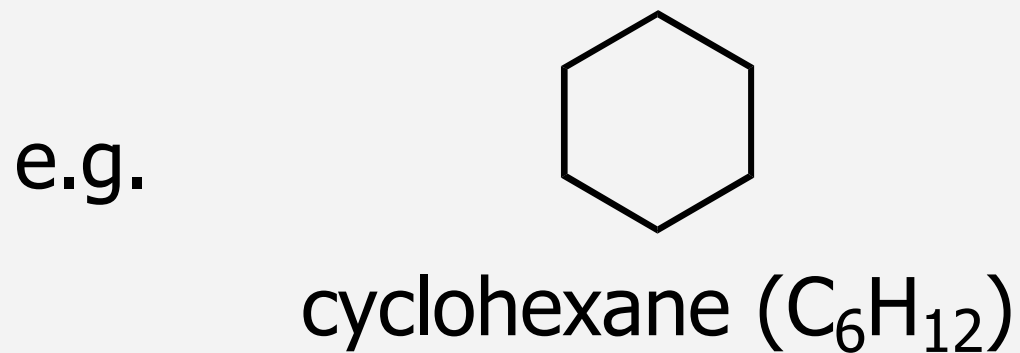
- They are hydrocarbons with a general molecular formula C_nH_{2n+2} where n = no. of carbon atoms
- They are made up of C – C bonds
- They are relatively unreactive
- The cycloalkanes have general formula C_nH_{2n}



❖ Alkanes: C_nH_{2n+2}



❖ Cycloalkanes: C_nH_{2n}



2. Shapes of Alkanes

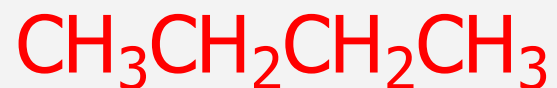
- ❖ All carbon atoms in alkanes and cycloalkanes are sp^3 hybridized, and they all have a **tetrahedral** geometry
- ❖ Even “straight-chain” alkanes are not straight. They have a zigzag geometry



“Straight-chain” (unbranched) alkanes



Butane

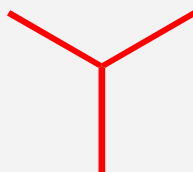
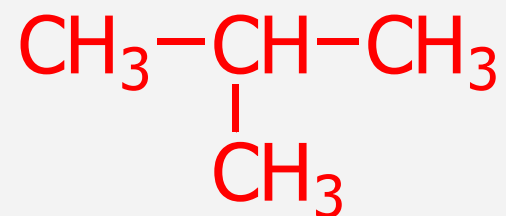


Pentane

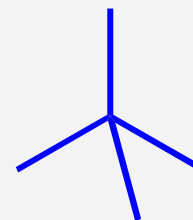
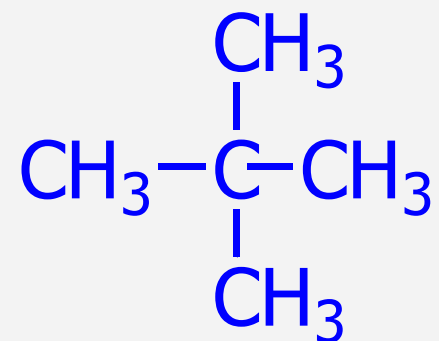


“Straight-chain” (unbranched) alkanes

Isobutane



Neopentane



3. IUPAC Nomenclature of ALKANES

- ❖ One of the most commonly used nomenclature systems that we use today is based on the system and rules developed by the *International Union of Pure and Applied Chemistry (IUPAC)*
- ❖ **Fundamental Principle:** Each different compound shall have a unique name



- ❖ Although the IUPAC naming system is now widely accepted among chemists, common names (trivial names) of some compounds are still widely used by chemists and in commerce.
- ❖ Thus, learning some of the common names of frequently used chemicals and compounds is still important



- ❖ The ending for all the names of alkanes is *-ane*
- ❖ The names of most alkanes stem from Greek and Latin

one	two	three	four	five
meth-	eth-	prop-	but-	pent-



<u>Name</u>	<u>Structure</u>	<u>Name</u>	<u>Structure</u>
Methane	CH_4	Hexane	$\text{CH}_3(\text{CH}_2)_4\text{CH}_3$
Ethane	CH_3CH_3	Heptane	$\text{CH}_3(\text{CH}_2)_5\text{CH}_3$
Propane	$\text{CH}_3\text{CH}_2\text{CH}_3$	Octane	$\text{CH}_3(\text{CH}_2)_6\text{CH}_3$
Butane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$	Nonane	$\text{CH}_3(\text{CH}_2)_7\text{CH}_3$
Pentane	$\text{CH}_3(\text{CH}_2)_3\text{CH}_3$	Decane	$\text{CH}_3(\text{CH}_2)_8\text{CH}_3$



Nomenclature of Unbranched Alkyl Groups

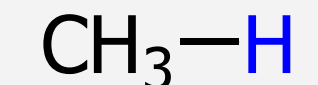
❖ Alkyl group

- Removal of one hydrogen atom from an alkane
- CH_3 – methyl
- C_2H_5 – ethyl
- C_3H_7 - propyl

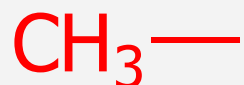


❖ Alkyl group

- For an unbranched alkane, the hydrogen atom that is removed is a **terminal** hydrogen atom



Methane



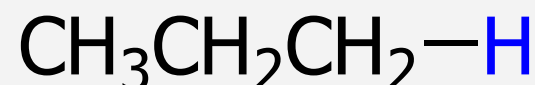
Methyl
(Me)



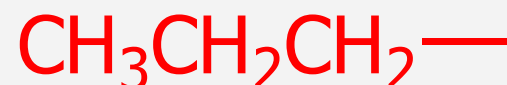
Ethane



Ethyl
(Et)



Propane



Propyl
(Pr)

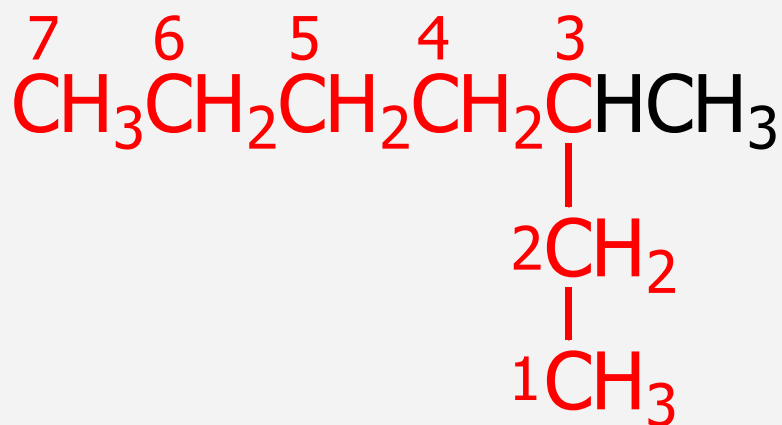


Nomenclature of Branched-Chain Alkanes

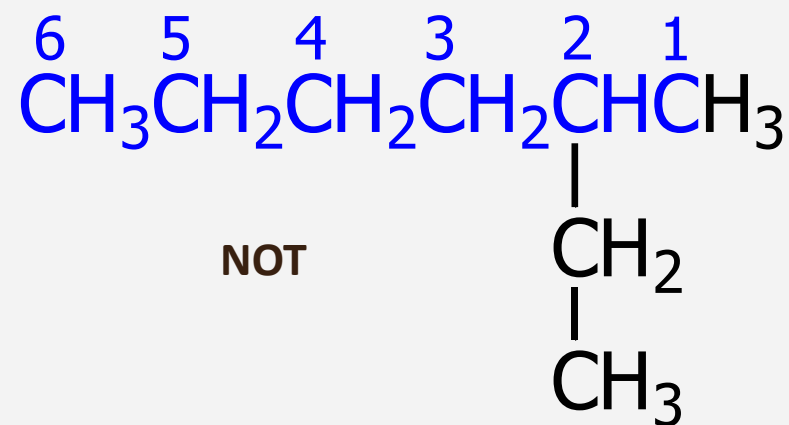


Rule

1. Use the longest continuous carbon chain as parent name



(3-Methylheptane)



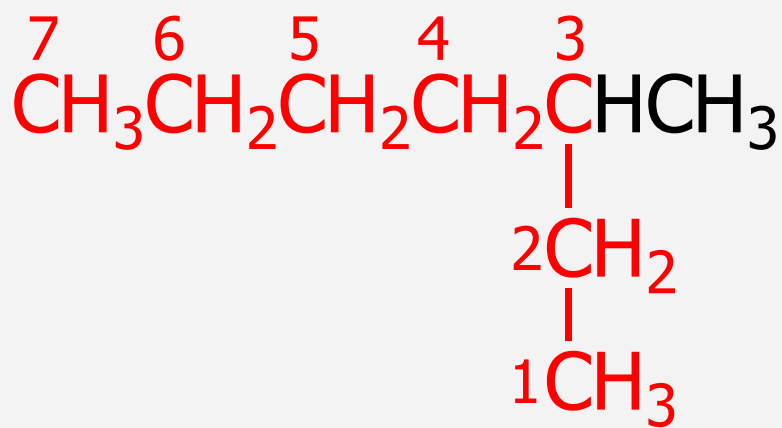
NOT

(2-Ethylhexane)

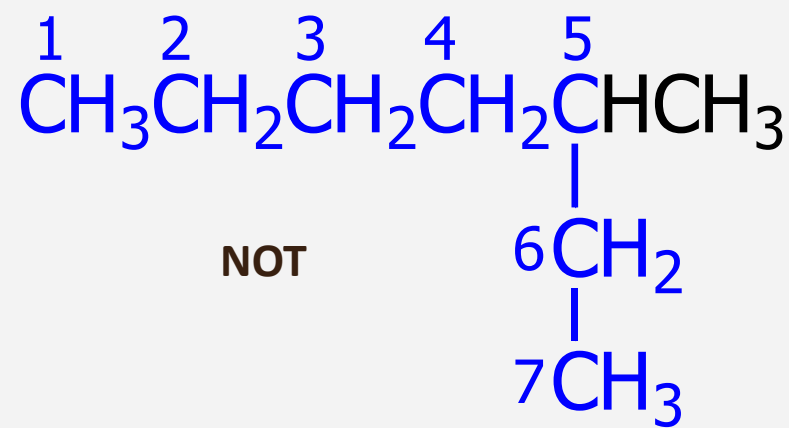


❖ Rule (Cont'd)

2. Use the lowest number of the substituent
3. Use the number obtained by Rule 2 to designate the location of the substituent



(3-Methylheptane)



NOT

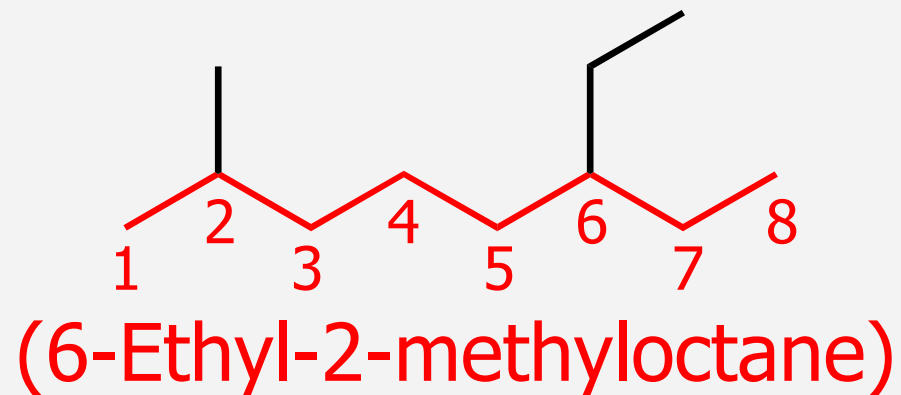
(5-Methylheptane)



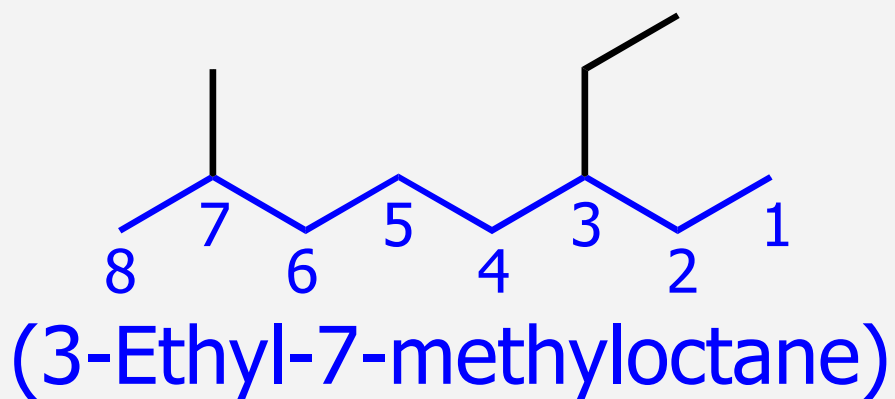
❖ Rule

4. For two or more substituents, use the **lowest possible individual numbers** of the parent chain. The substituents should be listed **alphabetically**. In deciding alphabetical order, disregard multiplying prefix, such as “di”, “tri” etc.

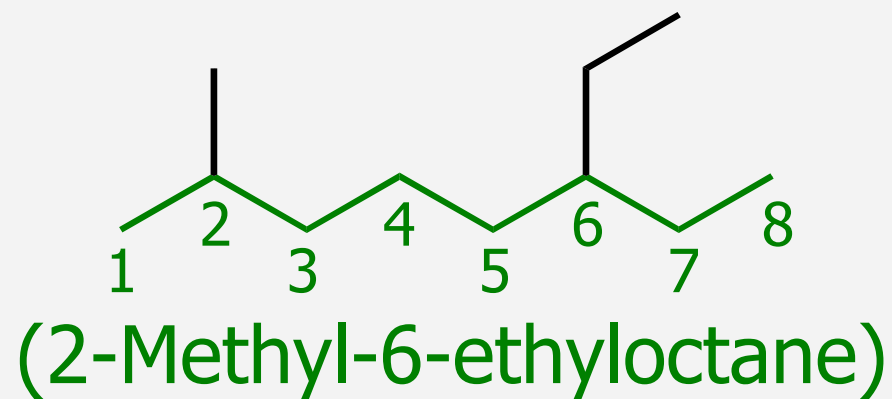




NOT

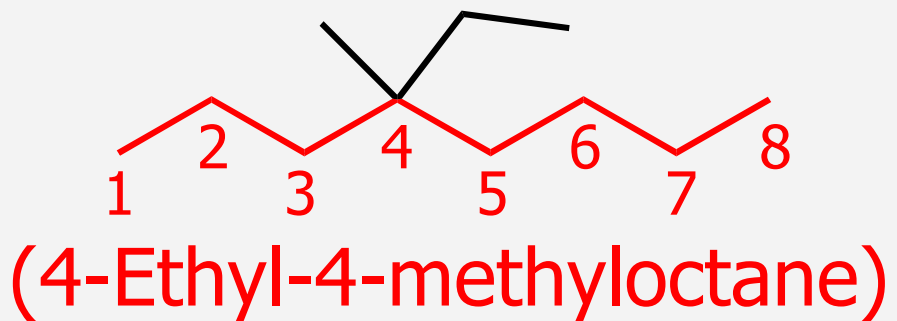


NOT

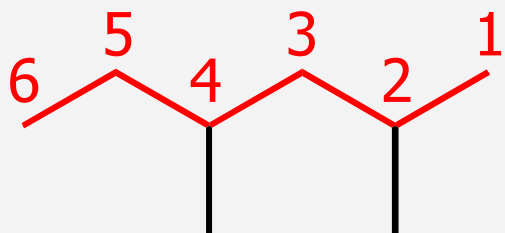




5. When two substituents are present on the same carbon, use that number twice

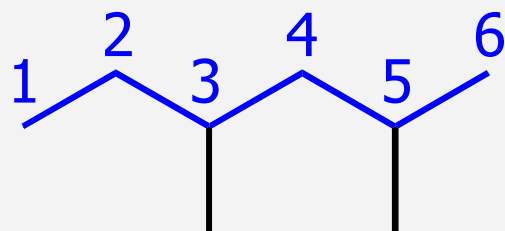


6. For identical substituents, use prefixes di-, tri-, tetra- and so on

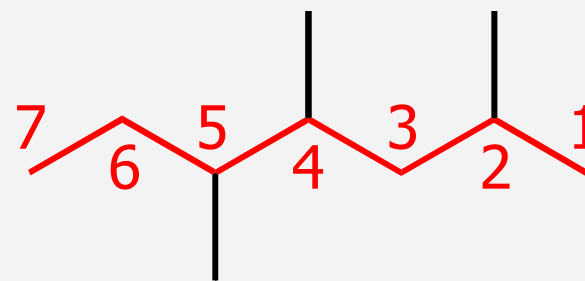


(2,4-Dimethylhexane)

NOT

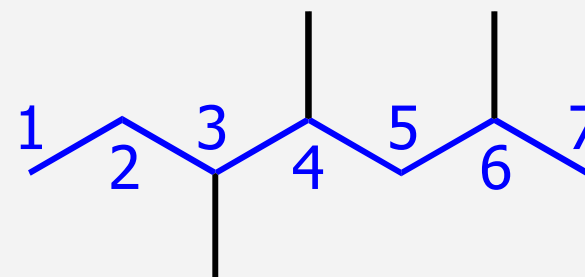


(3,5-Dimethylhexane)



(2,4,5-Trimethylheptane)

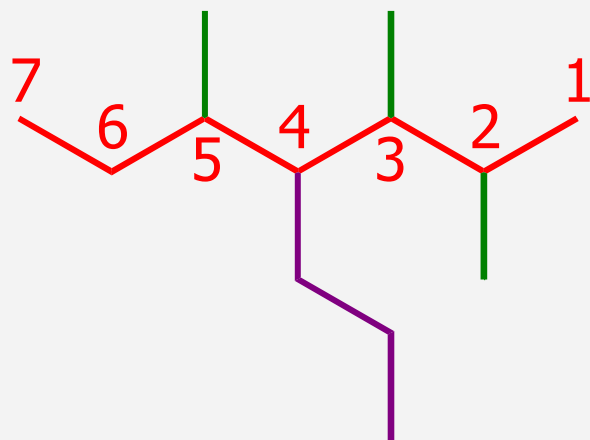
NOT



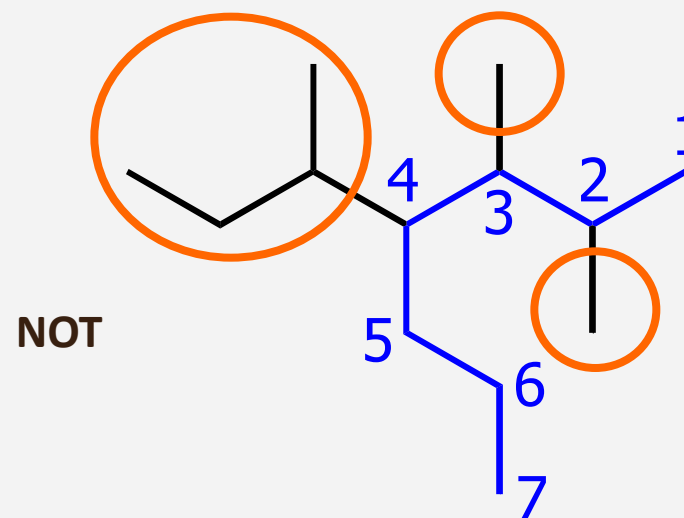
(3,4,6-Trimethylheptane)



7. When two chains of equal length compete for selection as parent chain, choose the chain with the greater number of substituents



(2,3,5-Trimethyl-
4-propylheptane)

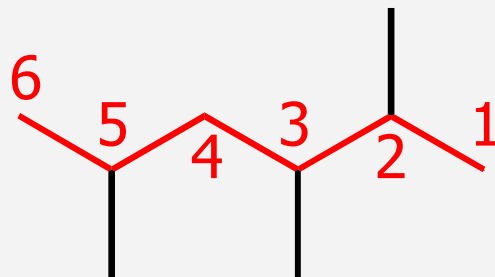


(only three substituents)

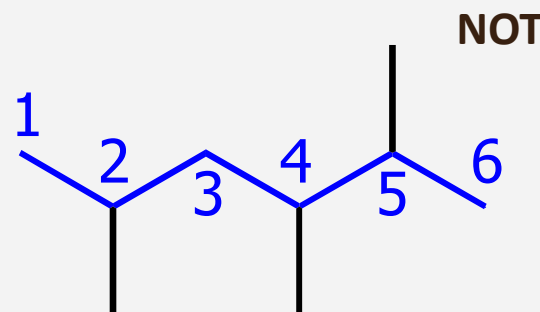




8. When branching first occurs at an equal distance from either end of the longest chain, choose the name that gives the lower number at the first point of difference



(2,3,5-Trimethylhexane)

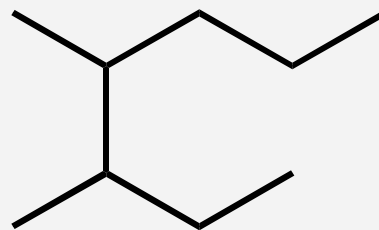


(2,4,5-Trimethylhexane)

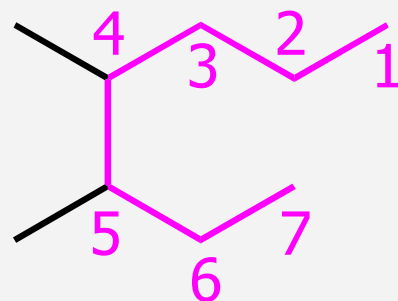




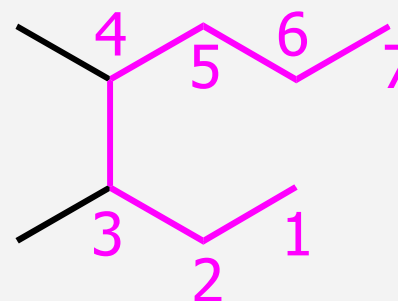
Example 1



- Find the longest chain as **parent**

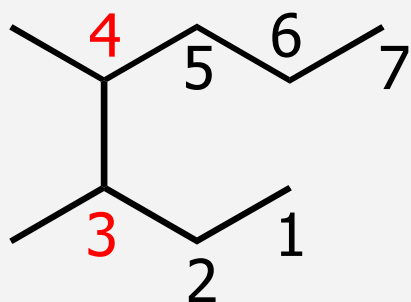


or

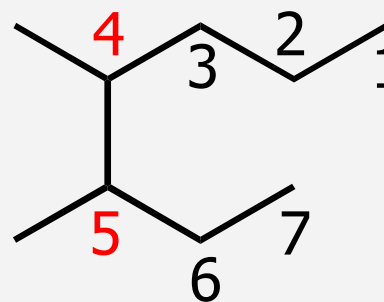




- Use the lowest numbering for substituents



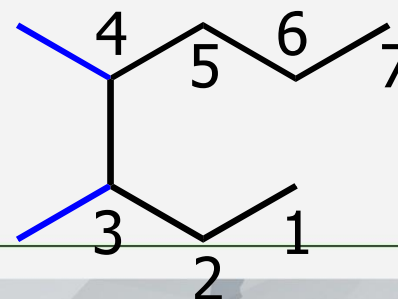
instead of



- Substituents: two methyl groups

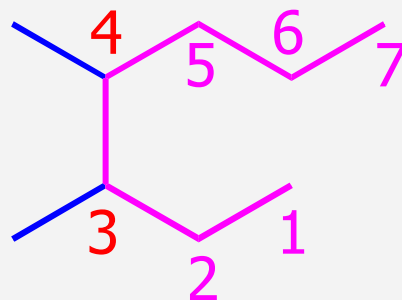


dimethyl





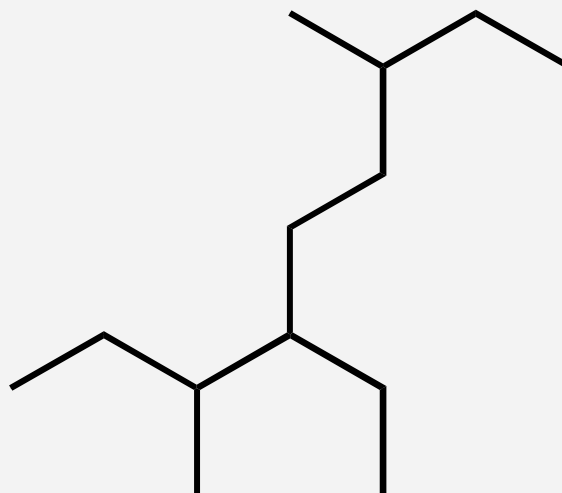
- Complete name



(3,4-Dimethylheptane)

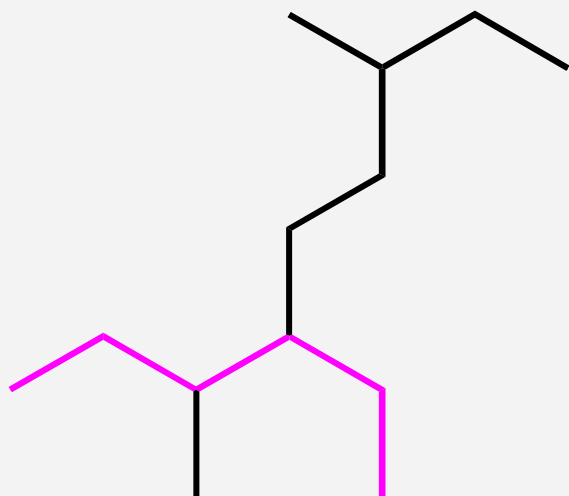


❖ Example 2

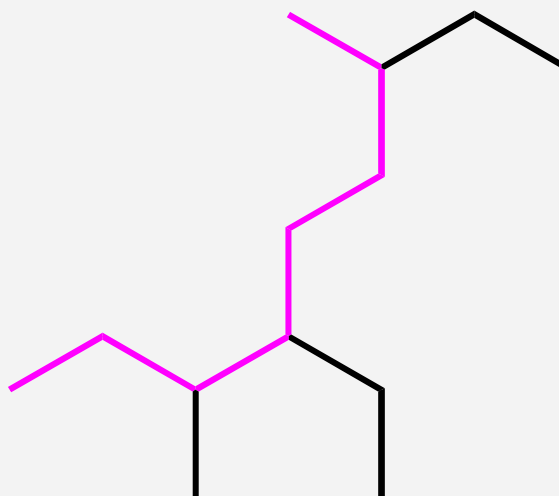


❖ Example 2 (Cont'd)

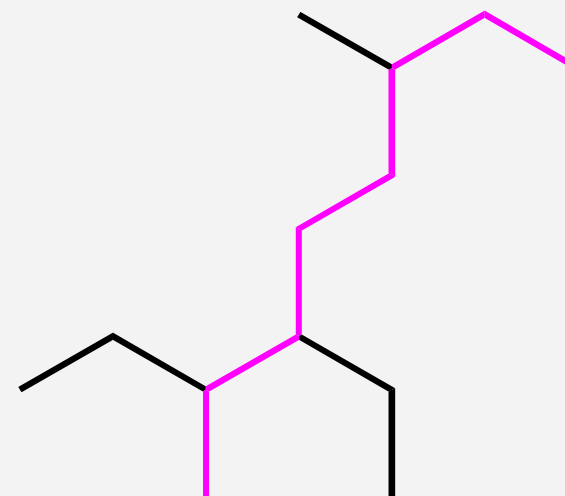
- Find the longest chain as **parent**



6-carbon chain



8-carbon chain

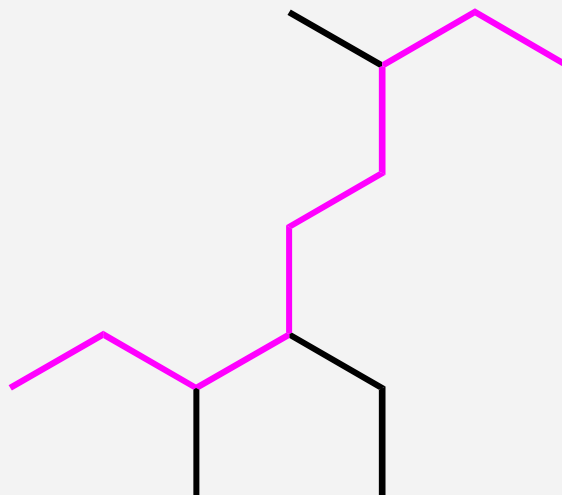


8-carbon chain



❖ Example 2 (Cont'd)

- Find the longest chain as **parent**

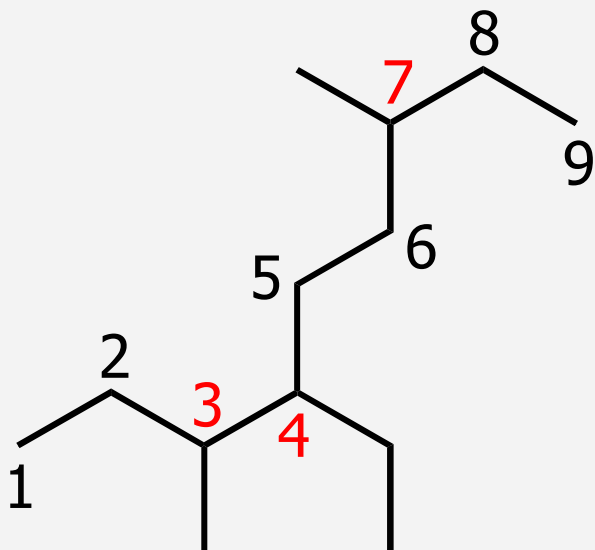


9-carbon chain
(correct!)

⇒ Nonane as parent

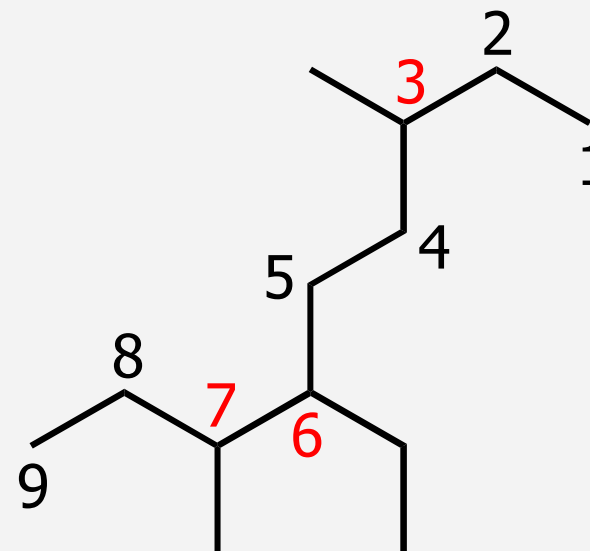


- Use the lowest numbering for substituents



(3,4,7)

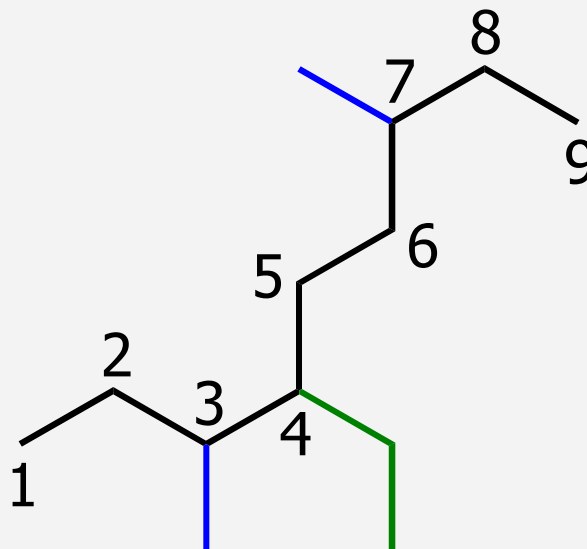
instead of



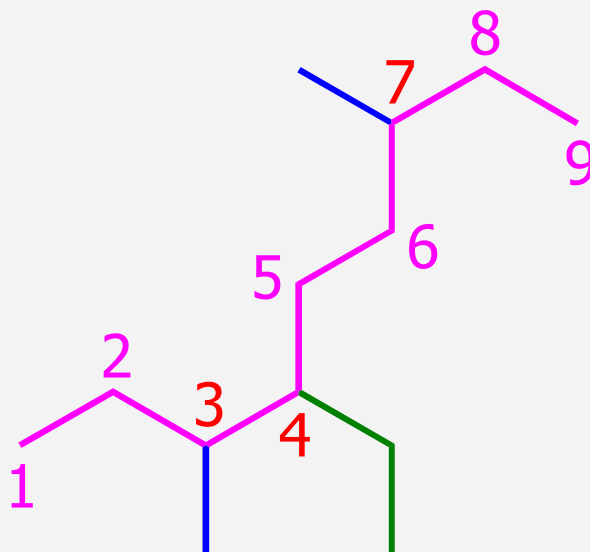
(3,6,7)



- Substituents
 - ◆ 3,7-dimethyl
 - ◆ 4-ethyl



- Substituents in alphabetical order
 - ◆ Ethyl before dimethyl
(recall Rule 4 – disregard “di”)
- Complete name

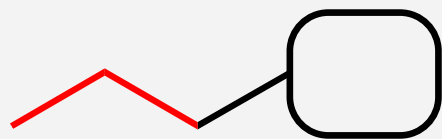


(4-Ethyl-3,7-dimethylnonane)

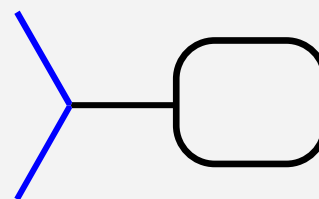


3C. Nomenclature of Branched Alkyl Groups

- ❖ For alkanes with more than two carbon atoms, more than one derived alkyl group is possible
- ❖ Three-carbon groups



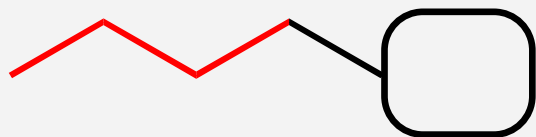
Propyl



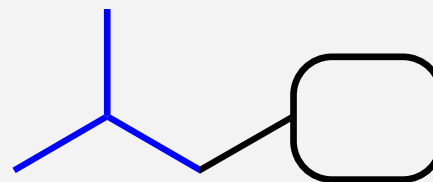
Isopropyl
(or 1-methylethyl)



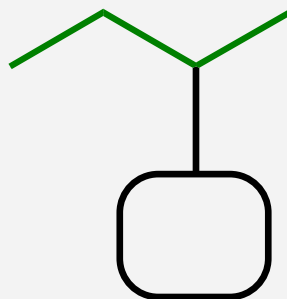
❖ Four-carbon groups



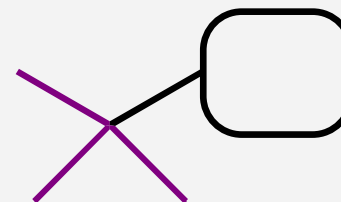
Butyl



Isobutyl



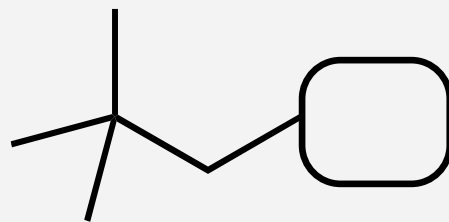
sec-butyl
(1-methylpropyl)



tert-butyl
(or 1,1-dimethylethyl)



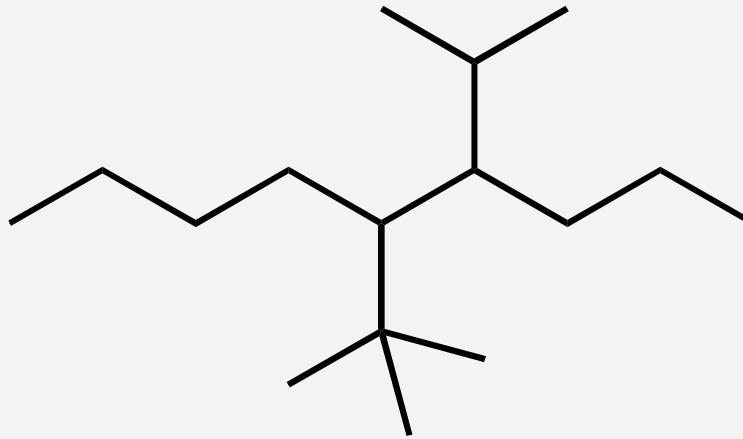
❖ A neopentyl group



neopentyl
(2,2,-dimethylpropyl)

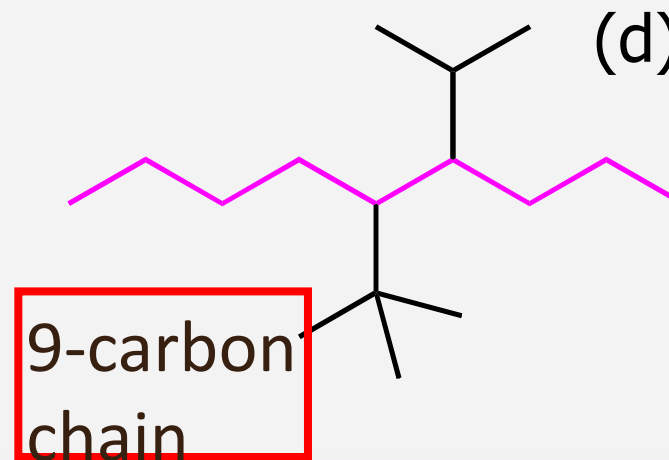
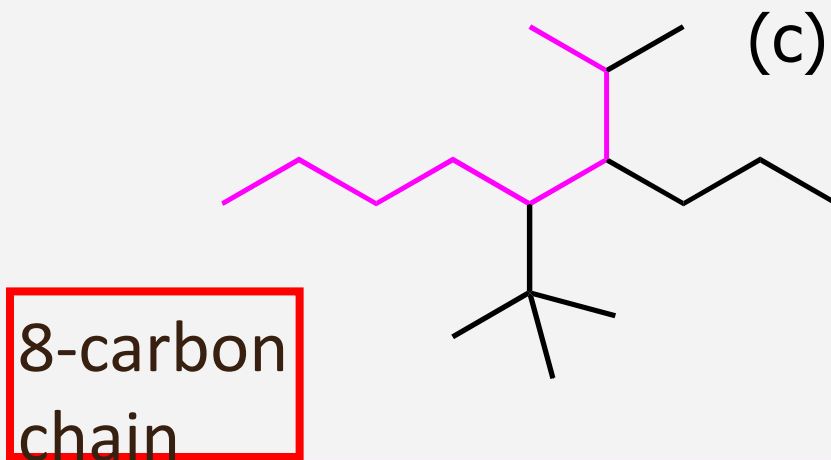
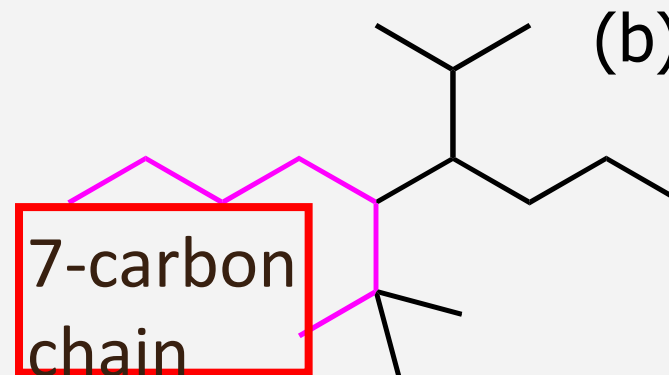
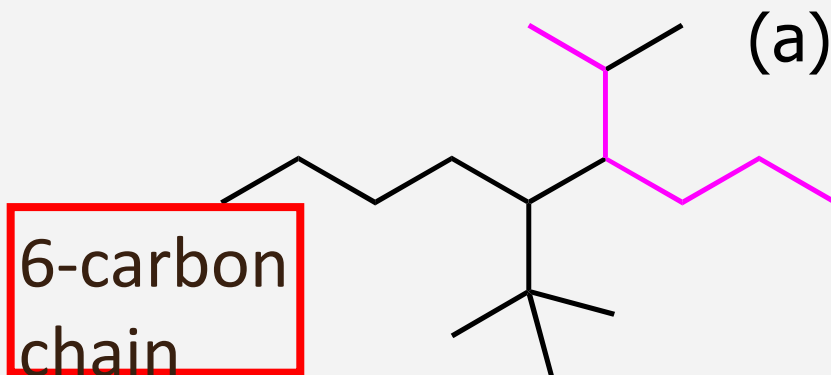


❖ Example 1



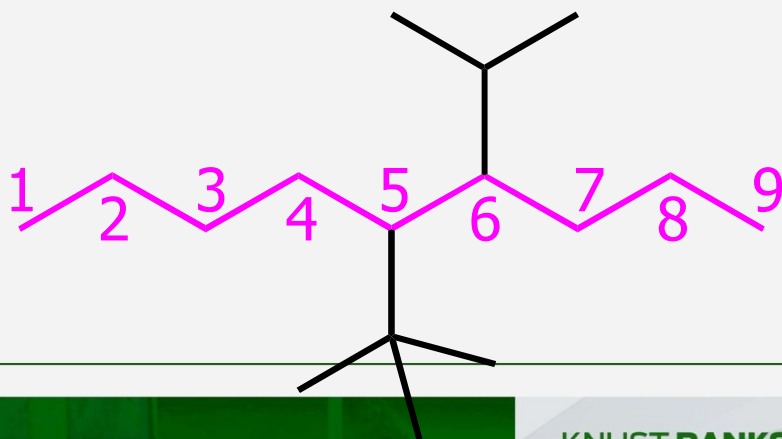
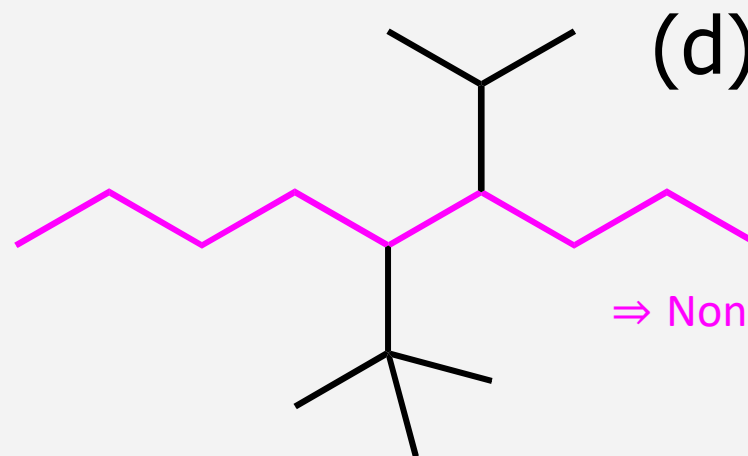
❖ Example 1 (Cont'd)

- Find the longest chain as **parent**

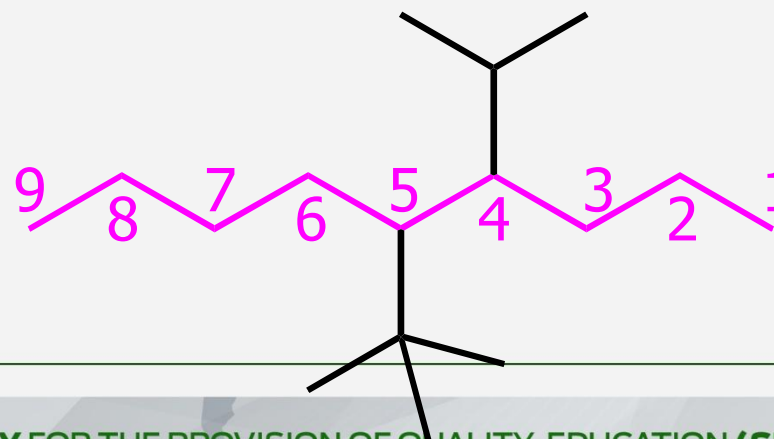


❖ Example 1 (Cont'd)

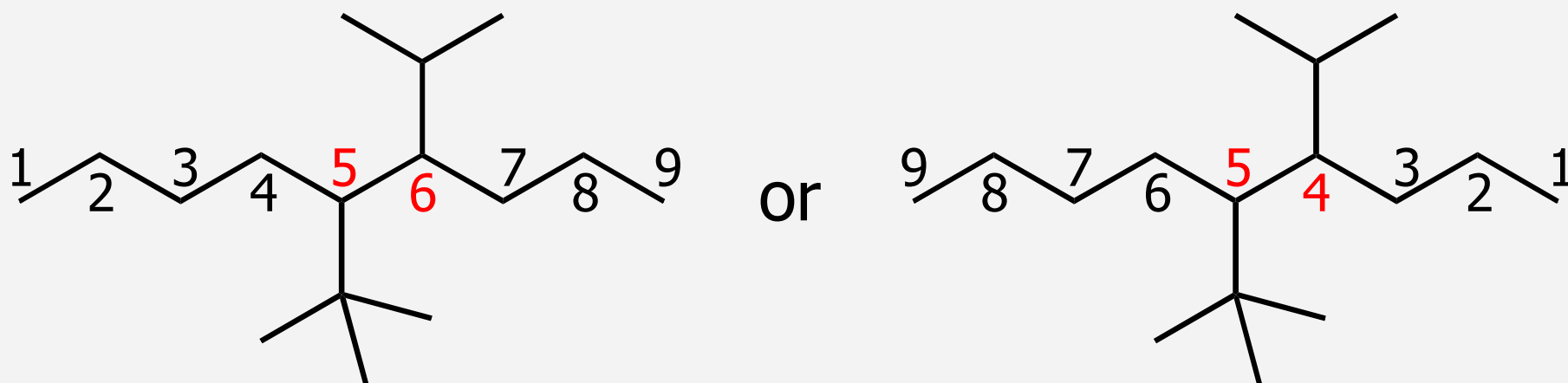
- Find the longest chain as **parent**



or



- Use the lowest numbering for substituents



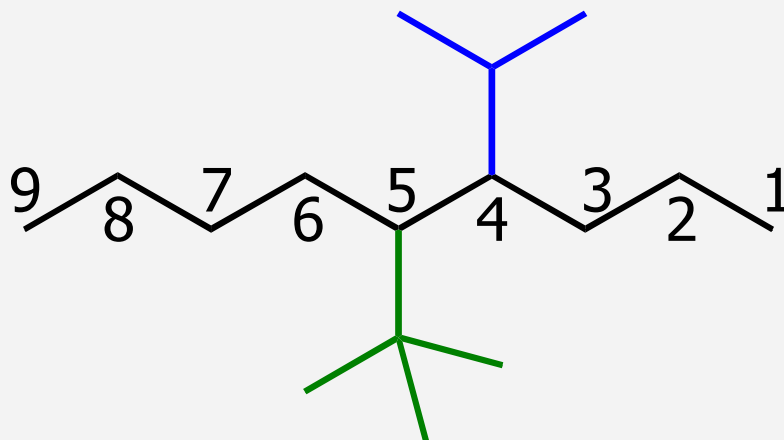
5,6

4,5
(lower numbering)

⇒ Use 4,5



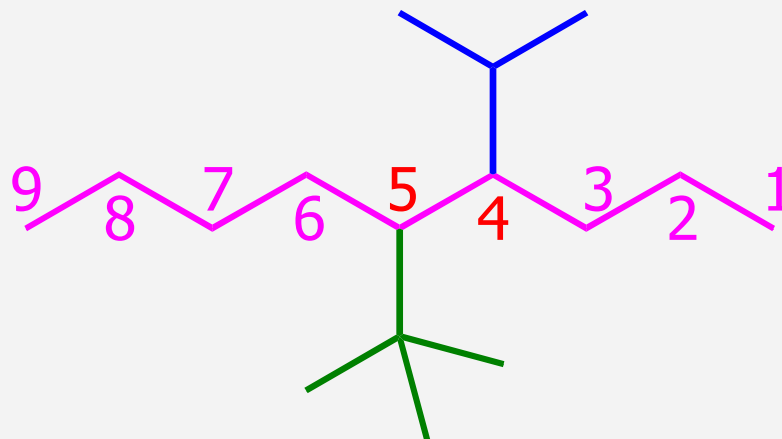
- Substituents
 - ◆ Isopropyl
 - ◆ *tert*-butyl



⇒ 4-isopropyl and 5-*tert*-butyl



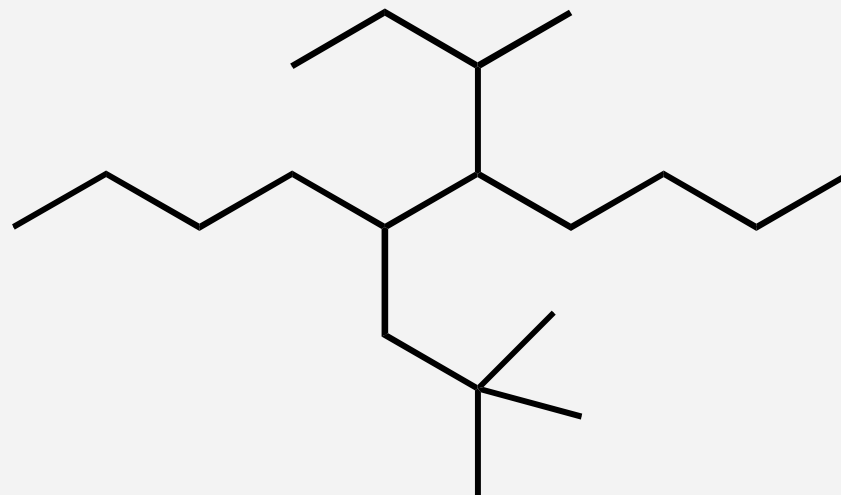
- Alphabetical order of substituents
 - ◆ *tert*-butyl before isopropyl
- Complete name



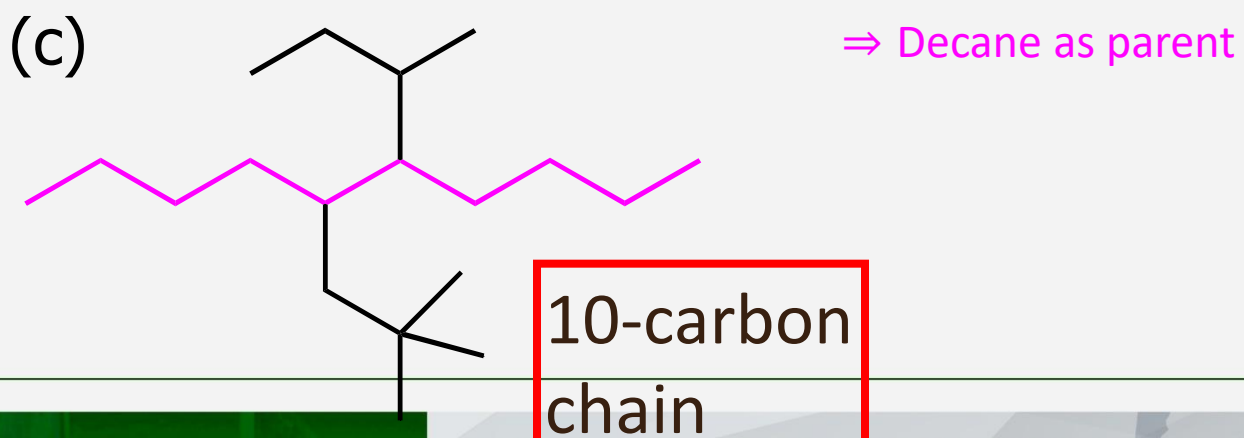
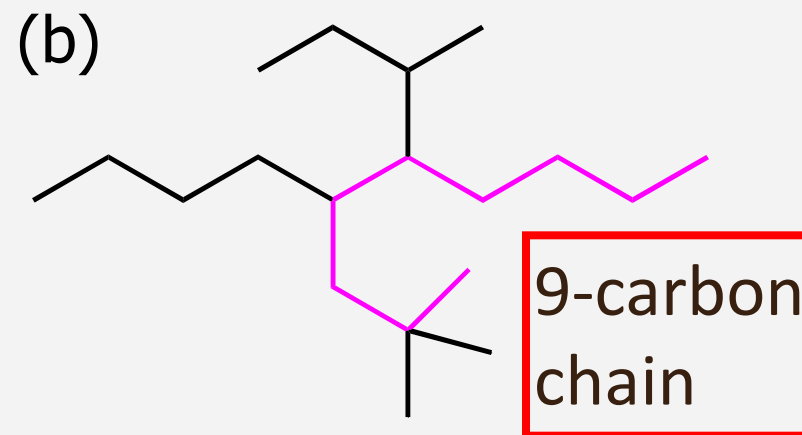
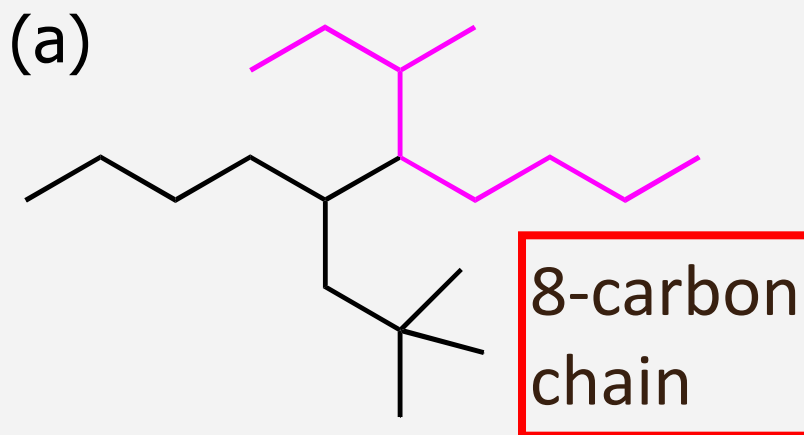
5-*tert*-Butyl-4-isopropylnonane

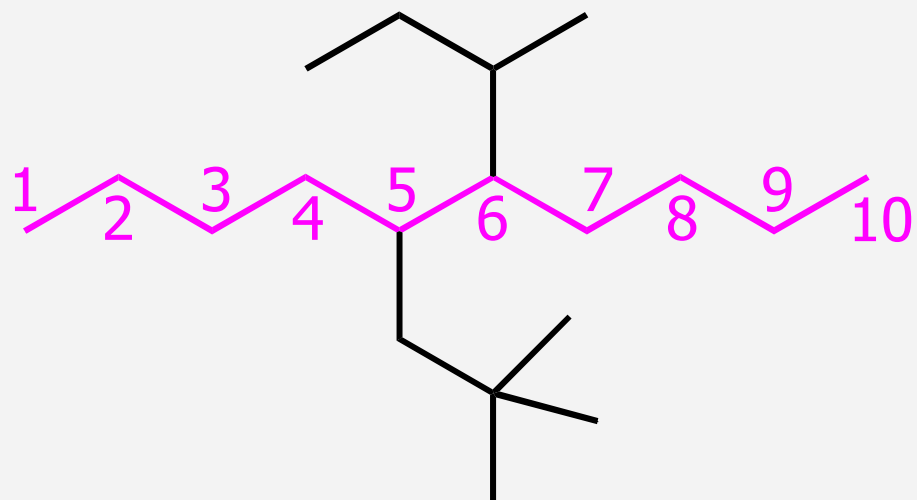


❖ Example 2

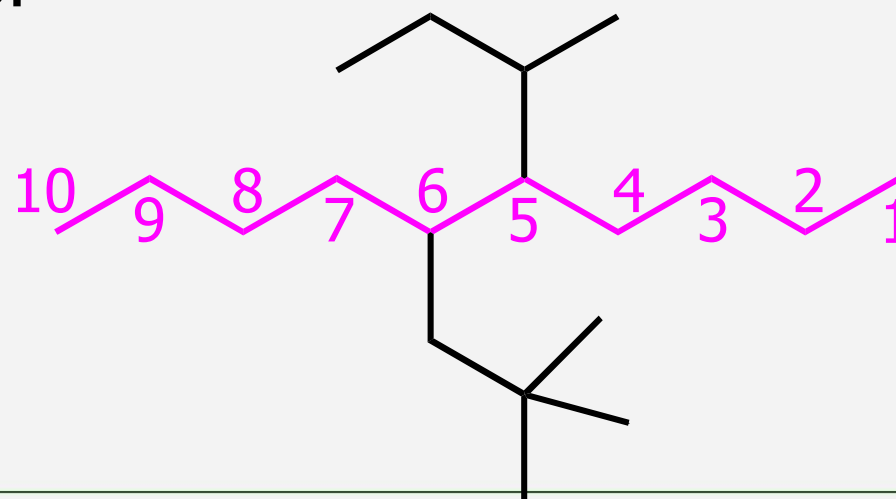


- Find the longest chain as **parent**

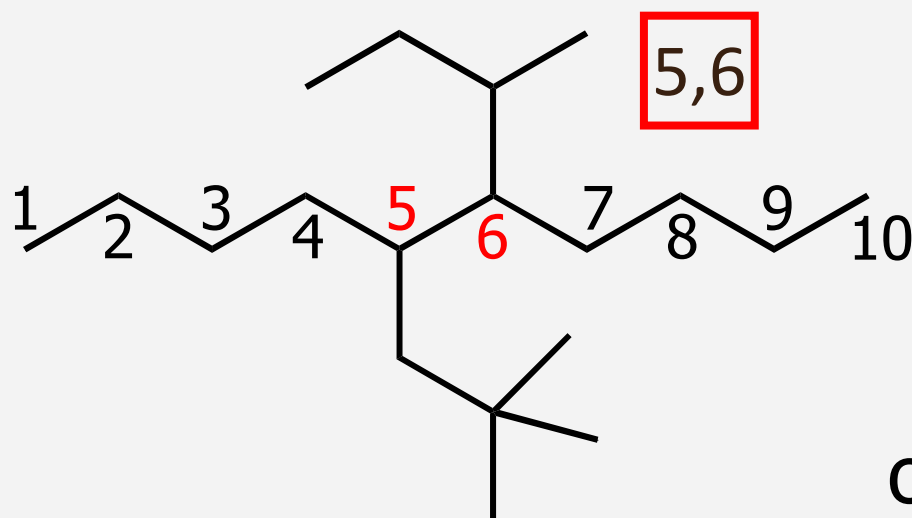




or

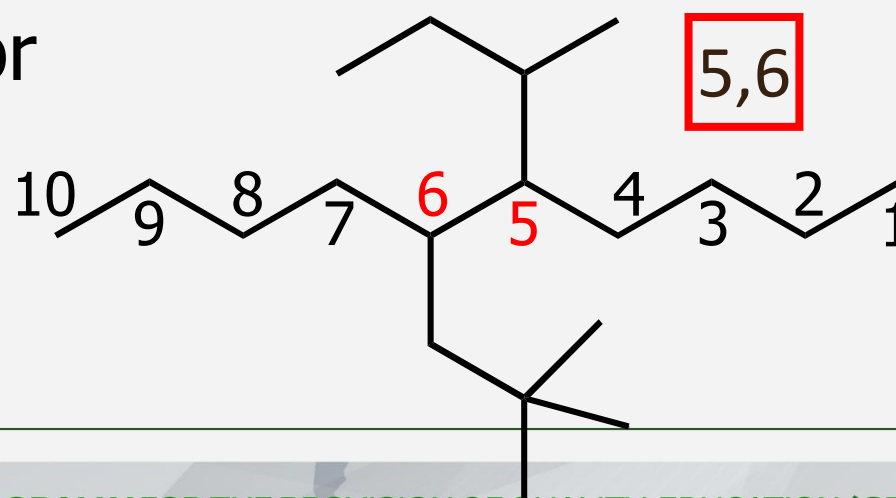


- Use the lowest numbering for substituents

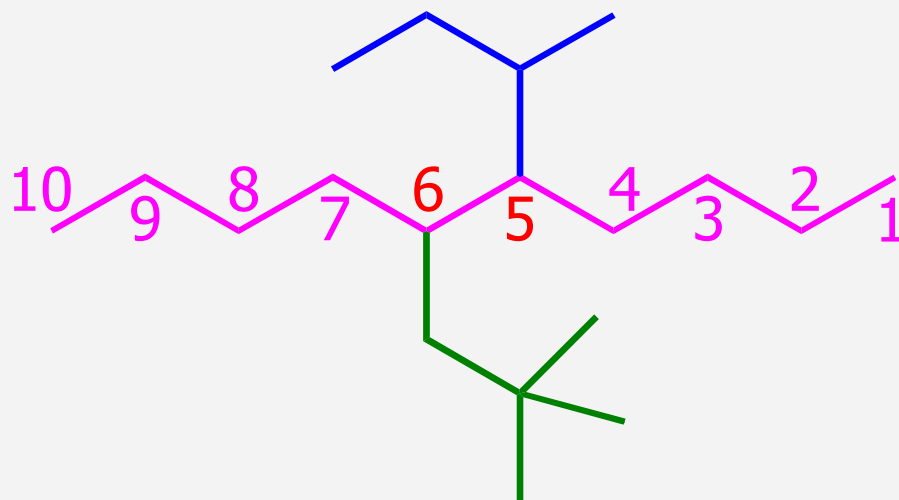


⇒ Determined using the next Rules

or



- Since *sec*-butyl takes precedence over neopentyl
 - ◆ 5-*sec*-butyl and 6-neopentyl
- Complete name

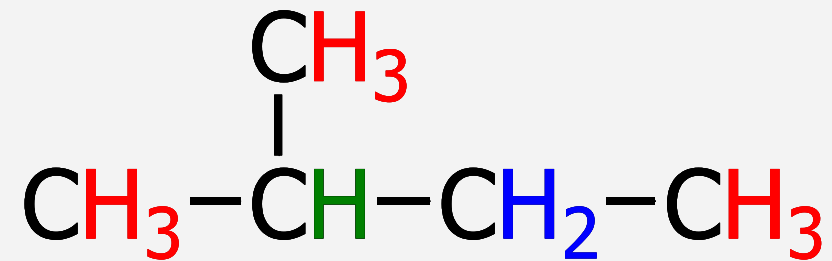


5-*sec*-Butyl-6-neopentyldecane



Classification of Hydrogen Atoms

1° hydrogen atoms



3° hydrogen atoms

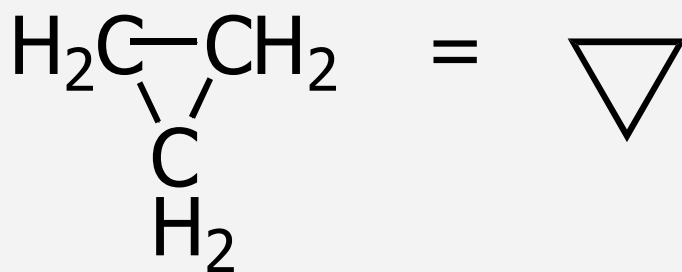
2° hydrogen atoms



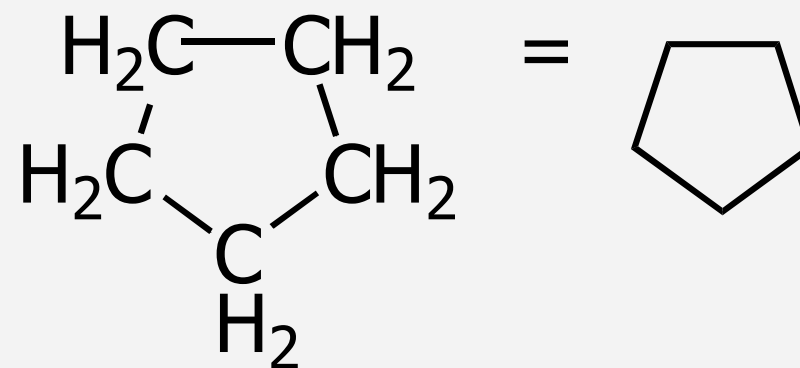
Naming Cycloalkanes

Monocyclic Compounds

- ❖ Cycloalkanes with only one ring
 - Attach the prefix **cyclo-**



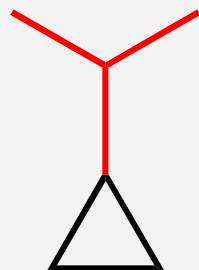
Cyclopropane



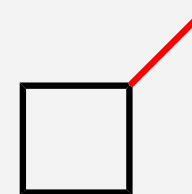
Cyclopentane



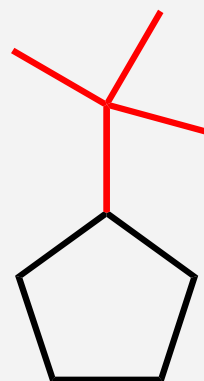
❖ Substituted cycloalkanes



Isopropylcyclopropane



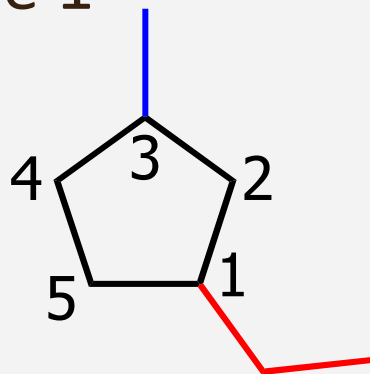
Methylcyclobutane



tert-Butylcyclopentane

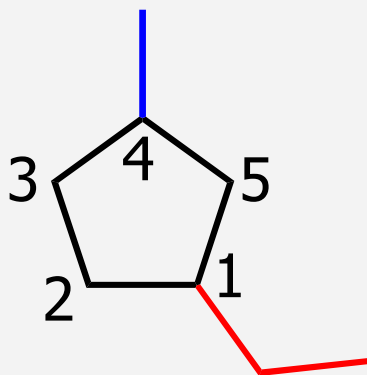


❖ Example 1



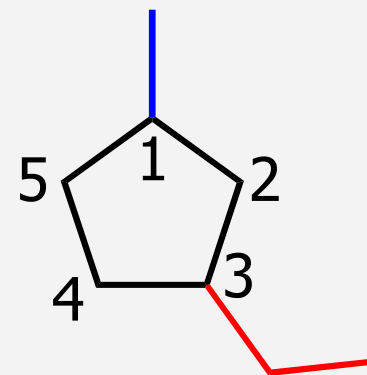
1-Ethyl-3-methyl-
cyclopentane

NOT



1-Ethyl-4-methyl-
cyclopentane

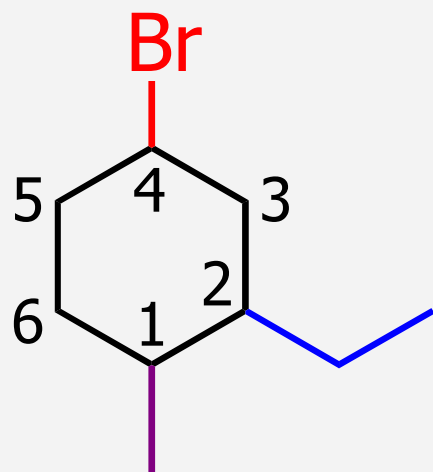
NOT



3-Ethyl-1-methyl-
cyclopentane

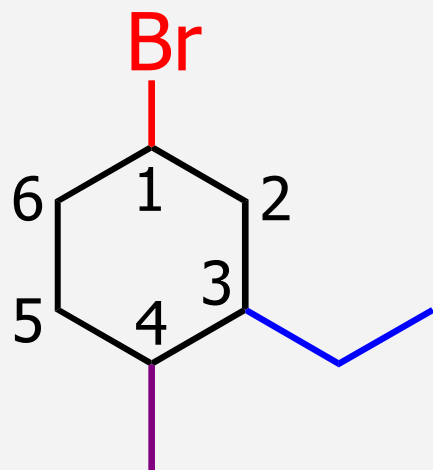


❖ Example 2



4-Bromo-2-ethyl-1-methyl
cyclohexane

NOT

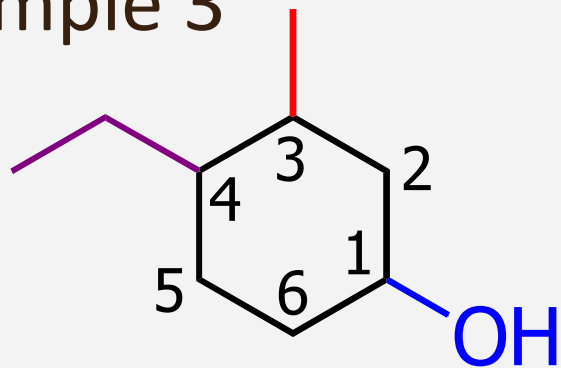


1-Bromo-3-ethyl-4-methyl
cyclohexane

(lowest numbers of substituents
are 1,2,4 not 1,3,4)

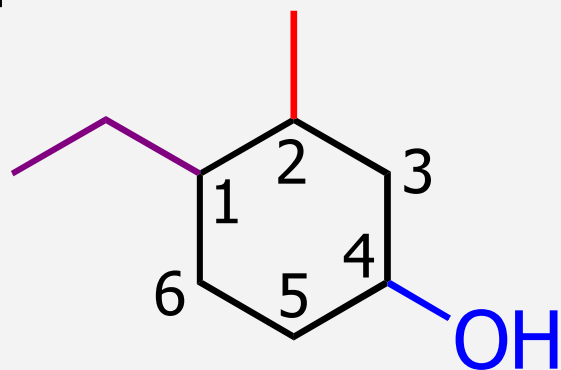


❖ Example 3



4-Ethyl-3-methyl
cyclohexanol

NOT



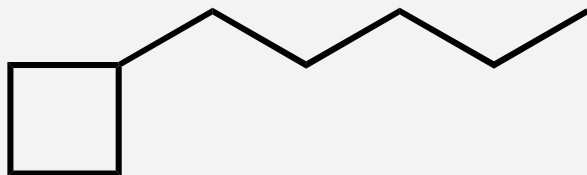
1-Ethyl-2-methyl
cyclohexan-4-ol

(the carbon bearing the OH should have the lowest numbering, even though 1,2,4 is lower than 1,3,4)



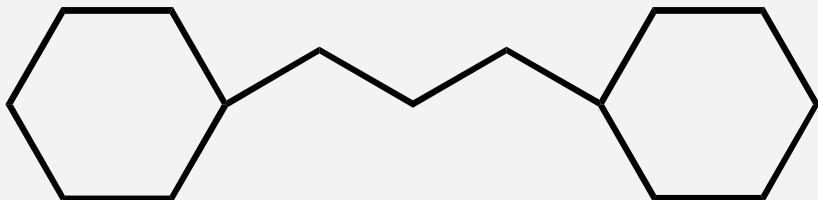
❖ Cycloalkylalkanes

- When a single ring system is attached to a single chain with a greater number of carbon atoms



1-Cyclobutylpentane

- When more than one ring system is attached to a single chain



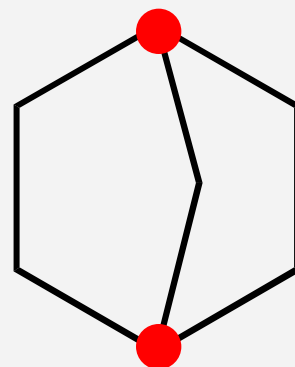
1,3-Dicyclohexylpropane



Bicyclic Alkanes

❖ Bicycloalkanes

- Alkanes containing two fused or bridged rings



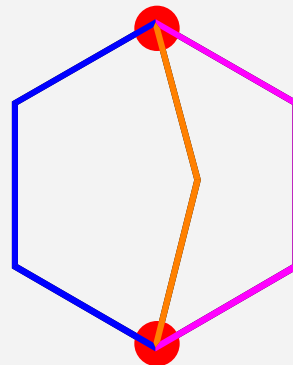
❖ Total # of carbons = 7

- Bicycloheptane

❖ Bridgehead



❖ Example (Cont'd)



❖ Between the two bridgeheads

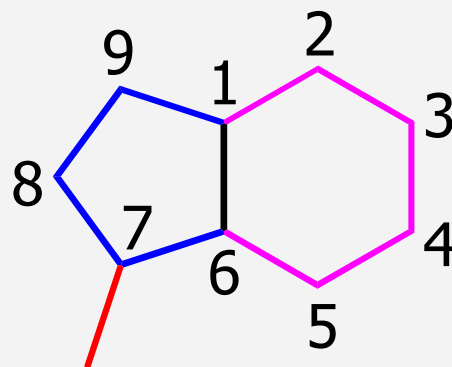
- Two-carbon bridge on the left
- Two-carbon bridge on the right
- One-carbon bridge in the middle

❖ Complete name

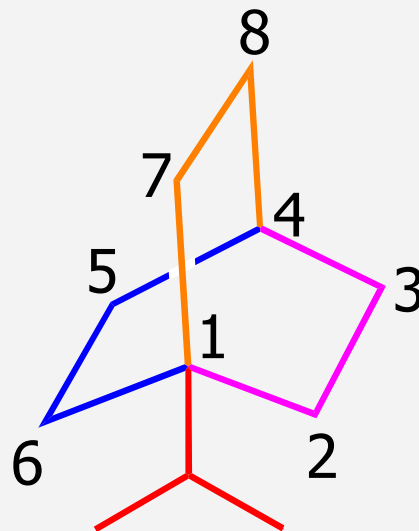
- Bicyclo[2.2.1]heptane



❖ Other examples



7-Methylbicyclo[4.3.0]nonane



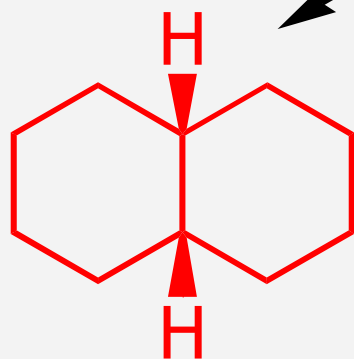
1-Isopropylbicyclo[2.2.2]octane



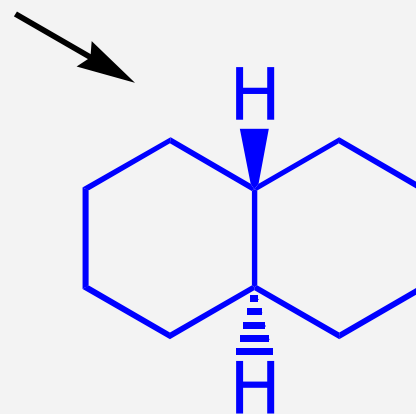
Bicyclic & Polycyclic Alkanes

Decalin

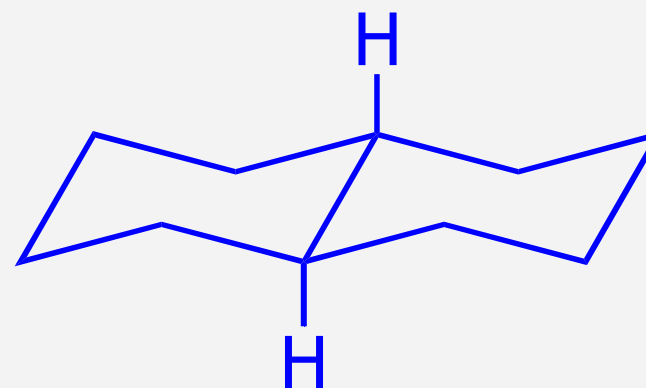
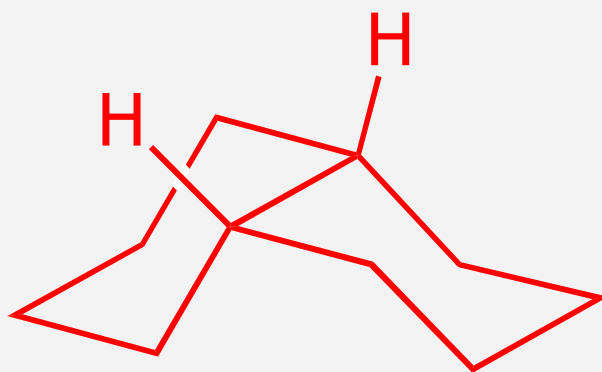
(Bicyclo[4.4.0]decane)

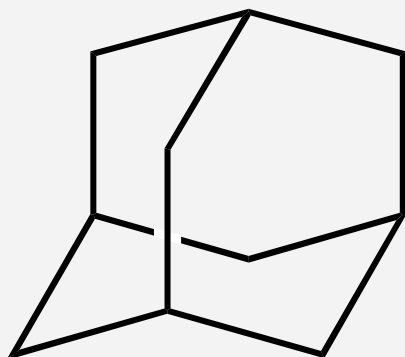


cis-Decalin

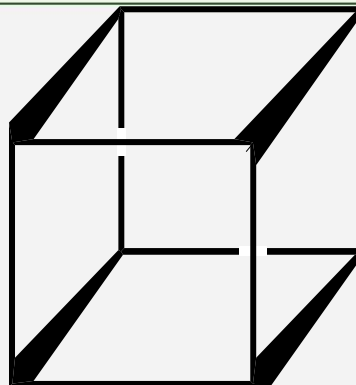


trans-Decalin

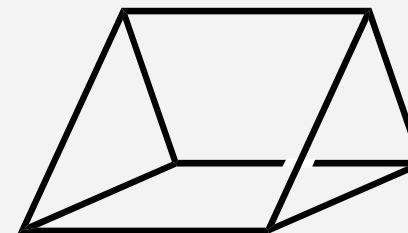




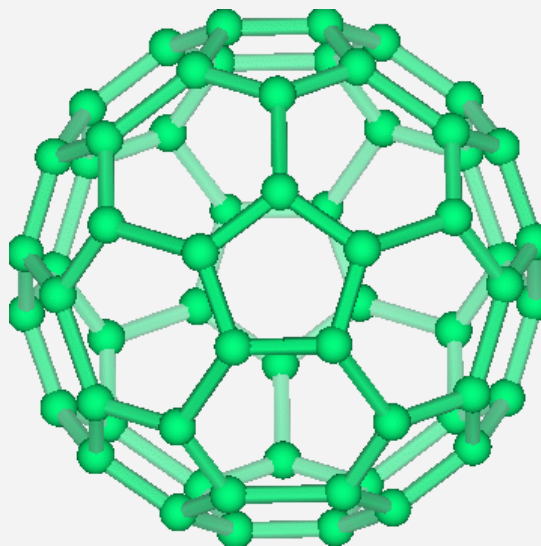
Adamantane



Cubane



Prismane

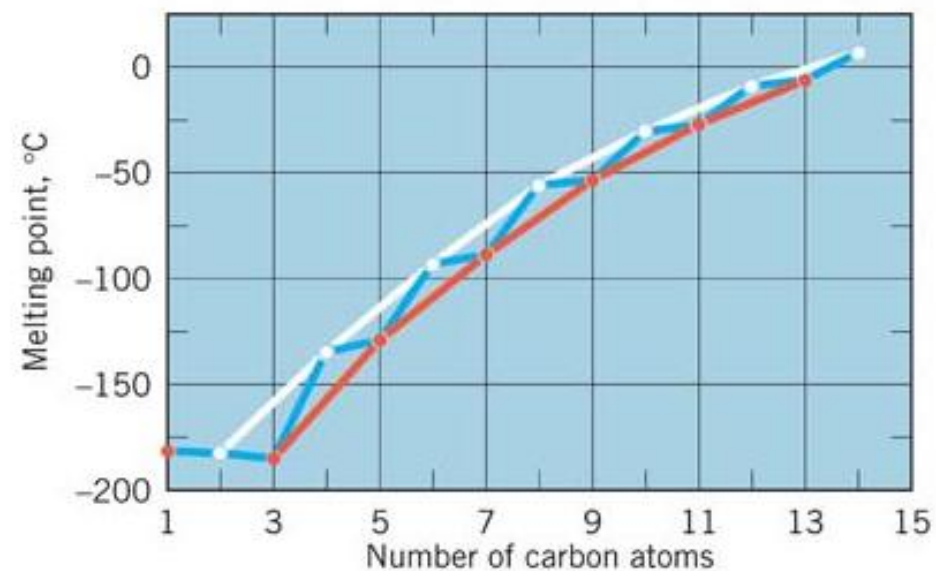
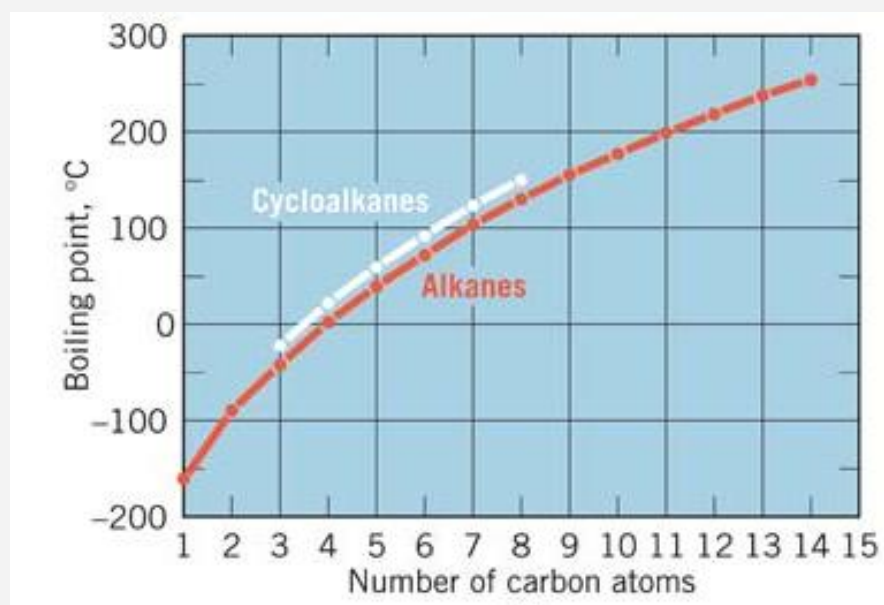


C_{60} (Buckminsterfullerene)



7. Physical Properties of Alkanes & Cycloalkanes

❖ Boiling points & melting points

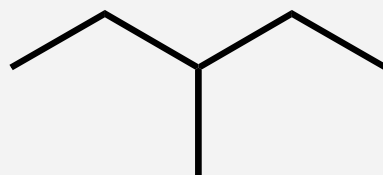


C_6H_{14} Isomer

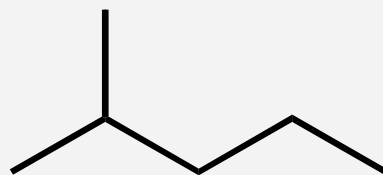
Boiling Point
(°C)



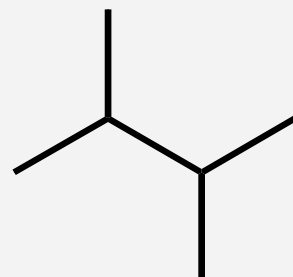
68.7



63.3



60.3



58



49.7



Physical Constants of Cycloalkanes

<i># of C Atoms</i>	<i>Name</i>	<i>bp (°C)</i>	<i>mp (°C)</i>	<i>Density</i>	<i>Refractive Index</i>
3	Cyclopropane	-33	-126.6	-	-
4	Cyclobutane	13	-90	-	1.4260
5	Cyclopentane	49	-94	0.751	1.4064
6	Cyclohexane	81	6.5	0.779	1.4266
7	Cycloheptane	118.5	-12	0.811	1.4449
8	Cyclooctane	149	13.5	0.834	-

